Example 7.4

A particle moves so that the distance moved is according to the law $s(t)=\frac{t^3}{3}-t^2+3$. At what time the velocity and acceleration are zero.

Solution

Distance moved in time 't' is $s = \frac{t^3}{3}-t^2+3$.

Velocity at time 't' is $v(t) = \frac{ds}{dt}=t^2-2t$.

Acceleration at time 't' is $a(t) = \frac{dV}{dt}=2t-2$.

Therefore, the velocity is zero when $t^2-2t=0$, that is $t=0,2$. The acceleration is zero when $2t-2=0$. That is at time $t=1$. ■

Example 7.5

A particle is fired straight up from the ground to reach a height of s feet in t seconds,where $s(t)=128t-16t^2$.

(i) Compute the maximum height of the particle reached.

(ii) What is the velocity when the particle hits the ground?

Solution

(i) At the maximum height, the velocity $v(t)$ of the particle is zero.

Now, we find the velocity of the particle at time t.

$v(t)=\frac{ds}{dt}=128-32t$

$v(t)=0 \Rightarrow 128-32t=0 \Rightarrow t=4$.

After 4 seconds, the particle reaches the maximum height.

The height at $t=4$ is $s(4)=128(4)-16(4)^2=256$ ft.

(ii) When the particle hits the ground then $s=0$.

$s=0 \Rightarrow 128t-16t^2=0$

$\Rightarrow t=0, 8$ seconds.

The particle hits the ground at $t=8$ seconds. The velocity when it hits the ground is $v(8)=-128$ ft/s. ■

Example 7.6

A particle moves along a horizontal line such that its position at any time $t \geq 0$ is given by $s(t)=t^3-6t^2+9t+1$, where s is measured in metres and t in seconds?

(i) At what time the particle is at rest?

(ii) At what time the particle changes its direction?

(iii) Find the total distance travelled by the particle in the first 2 seconds.

Solution

Given that $s(t)=t^3-6t^2+9t+1$. On differentiating, we get $v(t)=3t^2-12t+9$ and $a(t)=6t-12$.

(i) The particle is at rest when $v(t)=0$. Therefore, $v(t)=3(t-1)(t-3)=0$ gives $t=1$ and $t=3$.